

## Lab 2 — Sprint Plan, Estimation & Reflection

Restaurant Table Booking & Pre-Ordering App | Problem Statement #36

PES University — Dept. of CSE • Name: \_\_\_\_\_ SRN: \_\_\_\_\_ Date: \_\_\_\_\_

### 1. Sprint 1

**Sprint goal:** A diner can find a suitable table on the live floor plan, hold it, and browse the menu — with double booking made impossible.

Duration: 1 week • Committed: 34 story points • Completed: 29 • Carried over: 5 (Story 1.2)

| ID  | User story                             | Epic   | Priority | SP | End-of-sprint status         |
|-----|----------------------------------------|--------|----------|----|------------------------------|
| 1.1 | View live floor plan                   | Epic 1 | High     | 8  | Done                         |
| 1.2 | Filter tables by slot and party size   | Epic 1 | High     | 5  | Not completed → carried over |
| 1.3 | Select and hold a table                | Epic 1 | High     | 5  | Done                         |
| 1.4 | Reject overlapping reservations        | Epic 1 | Highest  | 8  | Done                         |
| 2.1 | Browse the digital menu                | Epic 2 | High     | 5  | Done                         |
| 2.4 | Show pre-order on the manager terminal | Epic 2 | High     | 3  | Done                         |

### 2. Sprint 2

**Sprint goal:** A diner can complete a paid pre-order end to end, and the system knows when the kitchen must start cooking.

Duration: 2 weeks • Committed: 32 story points • Completed: 32 • Story 1.2 was left in the product backlog rather than re-committed

| ID  | User story                                  | Epic   | Priority | SP | End-of-sprint status |
|-----|---------------------------------------------|--------|----------|----|----------------------|
| 2.2 | Add items to the pre-order cart             | Epic 2 | High     | 5  | Done                 |
| 2.3 | Edit or remove cart items                   | Epic 2 | Medium   | 3  | Done                 |
| 3.2 | Compute cook-start times                    | Epic 3 | High     | 8  | Done                 |
| 4.1 | Pay the pre-order through the gateway       | Epic 4 | Highest  | 8  | Done                 |
| 4.2 | Store payment tokens, never card data       | Epic 4 | Highest  | 5  | Done                 |
| 4.3 | Receive a digital confirmation with QR code | Epic 4 | High     | 3  | Done                 |

### 3. Velocity

| Sprint   | Duration | Committed | Completed | Achieved | Outcome                                                            |
|----------|----------|-----------|-----------|----------|--------------------------------------------------------------------|
| Sprint 1 | 1 week   | 34        | 29        | 85 %     | Story 1.2 (5 pts) not completed, returned to backlog               |
| Sprint 2 | 2 weeks  | 32        | 32        | 100 %    | All committed work accepted; Story 1.2 not pulled back in          |
| Average  | —        | 33        | 30.5      | —        | Working velocity ≈ 30 pts per sprint; 38 pts remain in the backlog |

### 4. Planning Poker record

Each story was discussed briefly, three estimators voted at the same time, and outliers explained their reasoning before a consensus number was agreed. The five stories below produced the widest spread.

| Story                                     | Votes      | Agreed | What the discussion surfaced                                                                                                                      |
|-------------------------------------------|------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1 View live floor plan                  | 8 / 5 / 13 | 8      | Rendering a live, colour-coded plan is more than a static image — the 13 vote flagged real-time refresh, which the team agreed sits in story 5.2. |
| 1.4 Reject overlapping reservations       | 5 / 8 / 8  | 8      | Concurrency control is the highest-risk item in the sprint; the team took the higher estimate because of uncertainty, not volume of code.         |
| 2.4 Show pre-order on manager terminal    | 3 / 3 / 5  | 3      | Read-only view over data that already exists — quick agreement.                                                                                   |
| 3.2 Compute cook-start times              | 5 / 8 / 13 | 8      | Spread came from edge cases (items sharing a station, overlapping bookings); parked as a spike rather than inflating the estimate to 13.          |
| 4.2 Store payment tokens, never card data | 3 / 5 / 8  | 5      | Little code, but compliance verification adds effort — the classic case for estimating uncertainty rather than hours.                             |

## 5. Burndown analysis

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See [Lab2\\_Burndown\\_Charts.pdf](#) for both sprints, and the Jira exports in screenshots/. The grey guideline is where the team should be assuming linear progress; the red line is the work actually remaining. Sprint 1 tracks close to the guideline through the first four stories, then flattens and stops at 5 points — Story 1.2 was never accepted. Sprint 2 steps down cleanly to zero. Because both sprints were run in one session, the Jira charts compress every step onto the first day; the sequence of values is still the real one.

## 6. Reflection

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### 1. Did your estimations reflect the actual effort?

Broadly yes, but not evenly. The two 8-point stories sized for risk rather than volume — 1.4 Reject overlapping reservations and 3.2 Compute cook-start times — were accepted without trouble, which suggests the team was right to price uncertainty rather than lines of code. The miss was Story 1.2 Filter tables by slot and party size, estimated at 5 and never finished: the estimate covered writing the filter, not wiring it to the live availability service. Sprint 2 estimates held completely — all 32 committed points were accepted — because they were re-based on what Sprint 1 actually delivered rather than on optimism.

### 2. Was your backlog well-prioritized?

The value ordering was sound: every Highest item (1.4 double-booking prevention, 4.1 payment, 4.2 tokenisation) was delivered inside the first two sprints, and the Low item (4.4 promo code) never competed with them. Two weaknesses showed up in practice. First, dependency ordering rather than value ordering: 5.2 Sync status changes in under 500 ms is still in the backlog even though 1.1 View live floor plan quietly depends on it. Second, and more visible in the sprint reports, the carried-over Story 1.2 was not pulled into Sprint 2 — unfinished work should normally be re-committed ahead of new work, otherwise a half-built feature sits in the backlog ageing while newer stories overtake it.

### 3. How did your simulated sprint align with your plan?

Sprint 1 committed 34 points and completed 29 (85 %), finishing with Story 1.2 still in progress. Sprint 2 committed 32 and completed all of it. Two deviations from the plan are worth recording honestly: Sprint 2 was timeboxed at two weeks rather than one, so its velocity is not directly comparable with Sprint 1; and both sprints were simulated inside a single working session, which is why the Jira burndown drops almost vertically against the calendar instead of stepping down day by day. The step sequence itself is accurate — 34 → 26 → 18 → 13 → 8 → 5 in Sprint 1 — because points only burn when a whole story is accepted.

### 4. What insights did the burndown chart give about your team's capacity?

Across the two sprints the team accepted 29 and then 32 points, so working capacity sits around 30 points, though Sprint 2 needed a two-week window to reach it — which is itself the most useful finding: the one-week timebox in Sprint 1 was the binding constraint, not the estimates. With 38 points left in the backlog (33 unplanned plus the 5-point Story 1.2), that is roughly one and a half more sprints of work. The step shape is the second insight: points arrive in 5- and 8-point jumps, a symptom of stories large enough to hide slippage until late. Splitting the remaining 8-point item (5.2) would make progress visible earlier. Neither sprint finished ahead of the guideline, so the team is not under-committing.